NEUROSCAN AI

Brain MRI Classification — Model Evaluation Report

1. Project Overview

NeuroScan AI is an educational deep learning project developed to explore brain MRI image classification using convolutional neural networks and transfer learning.

The system evaluates four dataset categories: glioma, meningioma, no tumor and pituitary.

2. Evaluation Dataset

Parameter	Details
Evaluation images	1,600
Number of categories	4
Images per category	400
Input resolution	224 × 224
Evaluation approach	Held-out test dataset

3. Model Performance Comparison

Model	Accuracy	Precision	Recall	Macro F1
CNN	63.56%	68.45%	63.56%	61.23%
MobileNetV2	**86.44%**	**87.51%**	**86.44%**	**86.07%**
Improved MobileNetV2	84.75%	85.63%	84.75%	84.26%

4. Best Performing Model

MobileNetV2 achieved the highest macro F1-score of **86.07%** and accuracy of **86.44%** among the evaluated models.

5. Evaluation Interpretation

The results show that transfer learning with MobileNetV2 performed better overall than the CNN baseline and the experimental focal-loss MobileNetV2 model on this test set. The comparison is based on the reported metrics and does not establish clinical effectiveness.

6. Limitations

The dataset is limited to four image categories. Performance on external datasets, different MRI scanners, image acquisition protocols and real clinical populations has not been established. The test dataset has already been used for model comparison, so further model tuning should use a separate validation dataset.

7. Medical Disclaimer

NeuroScan AI is an educational and research prototype only. It is not clinically validated, does not replace radiologists or medical professionals, and must not be used for diagnosis, treatment or clinical decision-making.